

Zihan Shao

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EDUCATION

Nanjing University

Bachelor of Science in Mathematics

GPA: 4.41/5.0

Nanjing, China

Sep. 2023-Jun. 2027(Expected)

Relevant Coursework: Mathematical Analysis (I):91, Higher Algebra (I) 96, Higher Algebra (II) 93, Analytic Geometry 96, Ordinary Differential Equations 90, Abstract Algebra 90, Functions of One Complex Variable 92.4, Discrete Mathematics 94, Foundations of Probability Theory 87, Introduction to Database 91, Numerical Analysis 98, Statistical Machine Learning 94, Multivariate Statistical Analysis 92

National University of Singapore

Science Summer Institute (Mathematics)

Singapore

Jul. 2025

PUBLICATIONS

Zihan Shao, Chang Lu, Renate A. Schmidt, Yizheng Zhao.

When Logic Meets Perception: Operator-Agnostic Differentiable Reasoning for Reliable Neural Prediction.

Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '26), August 9-13, 2026, Jeju Island, Republic of Korea.

<https://doi.org/10.1145/3770855.3817745>

RESEARCH EXPERIENCE

Research on Recurrent Latent Reasoning & Loop Transformer

New Haven, the United States

Advisor: Assistant Professor Zhuoran Yang

Jul.2026-Present

- Provides by the Zheng Gang Scholarship Program for outstanding and innovative students, selected through a joint evaluation by both on-campus and overseas advisors, to participate in overseas research
- Developed SLR, a recurrent latent-reasoning framework that compresses explicit chain-of-thought steps into fixed-size hidden states and supports depth-wise supervision and adaptive reasoning, using LOTUS as the primary baseline.

Research on Learned Spectral Preconditioning for Krylov Solvers

Nanjing, China

Advisor: Professor Tianlong Chen

Mar.2026-Present

- Proposed Learned Spectral PCG, a black-box spectral learning framework that learns positive Fourier-domain preconditioners instead of full optimization dynamics
- Constructed a lightweight 4,673-parameter MLP to predict SPD-compatible Fourier preconditioners from Hessian-vector probe features
- Proved that relative spectral inversion error controls the effective condition number and PCG convergence rate
- Second-authored paper submitted to NeurIPS 2026

Research on Minimax optimization on Vertical Federated Learning

Nanjing, China

Advisor: Associate Professor Siqi Zhang

Mar.2026-Present

- Designed two algorithms based on FedBCD for two VFL occasions and their corresponding error feedback versions
- Estimated the communication complexity of the two algorithms under SC-SC, NC-SC
- Carried out numerical experiments to verify the communication rounds and convergence

Research on Fuzzy Description Logic for Neural-Symbolic Integration

Nanjing, China

Advisor: Associate Professor Yizheng Zhao

Dec. 2024-Jun. 2026

- Proposed DF-ALC, the first end-to-end, operator-agnostic framework that turns fuzzy ALC from theory into a practical tool for representation learning
- Conducted the first systematic analysis of fuzzy t-norms/implications for ALC reasoning, demonstrating that no universally optimal operator exists
- Demonstrated via experiments that DF-ALC consistently outperforms established baselines
- First-authored paper accepted by ACM SIGKDD 2026 cycle 2 and selected for oral presentation

Research on Optimization Lower Bounds of Polyak–Łojasiewicz Functions

Nanjing, China

Advisor: Associate Professor Siqi Zhang

Oct. 2025-Mar.2026

- Investigated lower bounds for stochastic optimization of Polyak–Łojasiewicz (PL) functions
- Attempted to develop a new hard instance of PL-functions to address existing challenges

- Extended research fields to PL-PL, NC-PL minimax optimization problems and identifies key urgent research challenges in this direction

INTERNSHIP EXPERIENCE

Jiangsu Wanwei AIS Network Intelligent Industrial Innovation Center	Nanjing, China
Algorithm Intern	<i>Jun.-Aug.2025</i>
• Contributed to the development of a large AI model for oral health education • Designed and implemented interactive front-end and back-end interfaces by using LangChain, and Gradio in Python	

AWARDS AND HONORS

First-Class People’s Scholarship, Nanjing University	<i>Nov. 2025</i>
Third Prize, Final Round of the Chinese Mathematics Competitions for College Students	<i>Apr. 2025</i>
First Prize, Preliminary Round of the Chinese Mathematics Competitions for College Students	<i>Dec. 2024</i>
Second-Class People’s Scholarship, Nanjing University	<i>Nov. 2024</i>

LEADERSHIP AND EXTRACURRICULAR EXPERIENCE

Go Club, Nanjing University	Nanjing, China
Head of Instruction Department	<i>Sep. 2024-Sep. 2025</i>
• Led the club’s instructional initiatives, organized Go teaching sessions for club members	

TECHNICAL PROFICIENCIES

- Software: Python, C, C++, Java, Matlab, Geogebra, Protégé
- Languages: Chinese (Native), English (TOEFL: 110, R29, L29,S25, W27)